

SHOAIB SHAIKH

Senior Software Engineer - New York Metropolitan Area

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SUMMARY

Senior software engineer with 7+ years of deep domain expertise across the digital advertising stack – creative, targeting, forecasting, pacing, delivery, and attribution – building high-scale systems for Amazon’s device and marketplace advertising platforms. Track record of driving ambiguous, cross-team technical direction end-to-end, from emergency production migrations to multi-million-dollar cost, revenue, and compliance initiatives.

EXPERIENCE

Amazon – Amazon Ads

July 2025 – Present

Software Development Engineer II (Tech Lead)

New York, NY, USA

- Serving as the technical lead for advertising across the entire Kindle app and e-reader device business, owning roadmap end-to-end and directing engineering teams through legacy ad server migration and deprecation with zero service disruption
- Led delivery of European Accessibility Act (EAA) compliance across all Kindle ad placements, including a full legacy template refactor and gap-analysis review with accessibility, product, and legal teams
- Delivered a \$2M annualized revenue increase (20%+ delivery improvement) by architecting an emergency migration off a deprecated targeting service under a hard deadline, discovering and shipping a billable-ad-matching optimization along the way
- Led technical design and presented Kindle’s integration into Amazon’s cross-platform ad-pacing infrastructure at a multi-team design review, resolving engineering and management concerns from partner teams
- Designed and delivered an end-to-end customer-based personalization signal spanning new data pipelines, feature onboarding, and a new runtime fetcher across 5 services – unblocking a company-wide personalization initiative
- Oversaw execution of a Kindle-wide Java runtime migration – scoped which devices required it, authored the validation standard applied across the fleet, unblocked a peer engineer’s cutover change, and drove JDK upgrades across four additional dependent services to prevent them from blocking the effort
- Ran a cross-team research spike across 5 teams (forecasting, deals, publisher tech, targeting) to determine feasibility of a new Programmatic Guaranteed deal product on Kindle; findings became durable input to another team’s org-wide strategy document months later
- Drive technical cost-optimization for Books Ads hardware and infrastructure spend, identifying engineering-level reduction opportunities and partnering with fleet owners and leadership on budget decisions
- Led a multi-service migration off a hard-deprecating legacy backend across three production services, independently re-architecting the approach mid-flight after discovering a shared-dependency issue, with zero disruption and a 30% latency improvement
- Diagnosed a silent ad-delivery failure blocking a next-generation e-reader device’s launch, with no logs or errors pointing to a cause and no team yet assigned to it, through a multi-day investigation across app, device, and configuration layers – delivering the fix that unblocked the device’s launch
- Became the trusted cross-org point of contact for extending forecasting support to new ad products, driving requirements and implementation across 6 partner teams spanning Alexa, Fire Tablet, and Prime Video
- Root-caused a SEV-2 incident spanning two regions to a misconfigured internal throttle rule, drove cross-team escalation and resolution, and shipped a durable fix to prevent recurrence
- Provide ongoing ad campaign performance guidance to advertising sales leadership, investigating under-performing campaigns and publishing troubleshooting guides used directly by account teams
- Led a migration from RDS to DynamoDB prompted by a forced MySQL 5.7-to-8 upgrade that would have broken backend services if left unaddressed, cutting infrastructure costs 60% along the way
- Adopted agentic AI tooling into daily engineering practice – driving automated dead-code detection and cleanup across services and accelerating delivery on multiple migrations and onboardings
- Provided technical and career mentorship to 10+ engineers and interns, securing promotions, return offers, and improved performance ratings

Humber Polytechnic

January 2023 – August 2025

Professor

Toronto, ON, Canada

- Instructed core CS courses (Unix System Administration, Web Development, JavaScript, and Python) to 1000+ students, achieving strong engagement and learning outcomes across semesters

Amazon – Amazon Ads

Software Development Engineer II

October 2021 – July 2025

Toronto, ON, Canada

- Designed and led delivery of a new service to manage device ad-state migrations end-to-end, from design through production verification
- Built a general-purpose asynchronous, multi-threaded enrollment pipeline, increasing device processing throughput 10x; applied it to safely unenroll 12M devices from a legacy ad program, with headroom for larger-scale future migrations
- Spearheaded the launch of ad forecasting across Alexa, Fire Tablet, and Prime Video, enabling accurate impression prediction
- Refactored privacy-sensitive decade-old legacy systems under tight deadlines to achieve Digital Markets Act (DMA) compliance with zero downtime
- Led a database right-sizing initiative via blue-green deployments, cutting annual cloud costs 80% (\$1.2MM)
- Onboarded two new e-reader device lines to the ads platform on schedule despite significant technical obstacles, reverse-engineering deprecated ad-package generation tooling and resolving on-device memory issues along the way – generating \$2MM+ in annual revenue
- De-risked a production targeting-service migration by designing a multi-step shadow-call validation strategy, and authored the initial implementation that a peer engineer extended to complete the migration

Amazon – Amazon Ads

Software Development Engineer I

April 2020 – October 2021

Toronto, ON, Canada

- Architected and delivered the compatibility layer that decoupled Kindle’s decade-old legacy ad-serving platform from downstream consumers, unblocking its migration to a modern stack and laying the foundation later extended into a dedicated migration service
- Expanded advertising services from US-only to global marketplaces, enabling international scale
- Implemented React web components shown to 100K+ daily active users

The Citizen Lab – University of Toronto

Research Assistant

January 2020 – April 2020

Toronto, ON, Canada

- Developed ETL pipelines to collect, clean, and analyze large-scale online datasets for academic research

Amazon – Fulfillment by Amazon (FBA)

Software Development Engineer Intern

May 2019 – August 2019

Toronto, ON, Canada

- Reduced API latency from 500ms to 10ms by eliminating AWS Lambda cold starts via S3-backed caching
- Improved storage performance by regionalizing S3 buckets and optimizing integration tests
- Built internal full-stack debugging tool to replace manual log inspection, slashing developer troubleshooting time

IBM

Software Developer Intern

May 2018 – May 2019

Markham, ON, Canada

- Reduced automated end-to-end test runtime by 90% via a fail-fast testing strategy and improved CI/CD visibility with automated Slack/email reporting tools

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science (Specialization in Computing Systems)

Atlanta, GA, USA

September 2026 – Present

University of Toronto

Honours B.Sc. in Computer Science (Specialist Focus in Artificial Intelligence), Minor in Economics

Toronto, ON, Canada

September 2015 – April 2020

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, C/C++, HTML/CSS, Kotlin, PHP, Ruby, SQL

Cloud: AWS(Lambda, DynamoDB, S3, SQS, SNS, RDS, EC2, ECS, Kinesis, Bedrock, IAM, VPC, Route53, etc), Azure, Firebase

Frameworks/Runtimes: Node.js, Express, React, Next.js, Spring Boot, Vue, Angular, Shadcn, Tailwind CSS

AI/ML & Data: Generative AI (LLMs, Agentic AI), OpenAI/Claude/Gemini/Grok APIs, PyTorch, TensorFlow, pandas, NumPy, scikit-learn

Tools & Platforms: Docker, Git, Jenkins, CI/CD, Supabase, Firebase, Stripe, Azure, Cursor

Concepts: Digital Advertising, Distributed Systems, Microservices, API Design, Operational Excellence, Technical Debt Reduction